



PyCon JP 2026

How Python helped me getting the JLPT N1?

From dattebayo! to Leading an Engineering Team in Tokyo

Jose

SODA inc



**How many of you here today
speak or understand
Japanese?**

2016 (My only vocabulary)



俺 "ore"

Translation: "I" (Ultra-masculine)

⚠️ USE WITH BOSS AT YOUR OWN RISK

だってばよ "dattebayo"

Translation: "Believe it!" (Naruto)

🌀 NOT A REAL WORD, BELIEVE IT!

こんにちは "konnichiwa"

Translation: "Hello" (Normal word)

✅ SAFE TO USE IN SOCIETY

お兄ちゃん "onichan"

Translation: "Big brother" (Cringe at work)

💀 MAXIMUM SOCIAL AWKWARDNESS

2024 (The top)

2024 年 7 月 7 日に、公益財団法人日本国際教育支援協会及び独立行政法人国際交流基金が実施した日本語能力試験に関し、認定結果及び成績を次のとおり証明します。

This is to certify the result and the scores of Japanese-Language Proficiency Test given on July 7, 2024 jointly administered by Japan Educational Exchanges and Services and the Japan Foundation.

発行日 Date of Issue (y/m/d)	2024/09/03
受験番号 Registration No.	
氏名 Name	JUANJOSE HAFID TENORIO PENIA
生年月日 Date of Birth (y/m/d)	1994/11/26
受験地 Test Site	日本 Japan
レベル Level	N1
結果 Result	合格 Passed
認定番号 Certification No.	





**How I did it? And how did
Python helped me?**

Is English Really Enough?

"The 'official' language for IT engineers is English, right?"

Well, yes, but not quite like that here.

The Reality of English-Only Tech Roles



A Highly Concentrated Ecosystem

The market for pure English-speaking engineering roles in Japan is remarkably tight.

300–500

Total Companies

Total estimated firms in Japan accommodating English-only tech talent.

150–250

Active Startups

Startups actively hiring, heavily driven by Deep Tech and AI sectors.

Severe Logistical Friction

Despite an eagerness to onboard international talent, relocating engineers from overseas presents immense bureaucratic risks and high upfront costs.

40%

Of Job Postings

require that an applicant might already be in Japan

Not Convinced?

9.1 Work Visas

New JLPT N2 Requirements

Effective **April 15, 2026**, first-time overseas applicants must prove proficiency equivalent to **JLPT N2 (or CEFR B2)**.

This rule targets applicants at **Category 3 and 4 companies** (SMEs and startups) to prevent visa abuse..

9.2 Permanent Residency

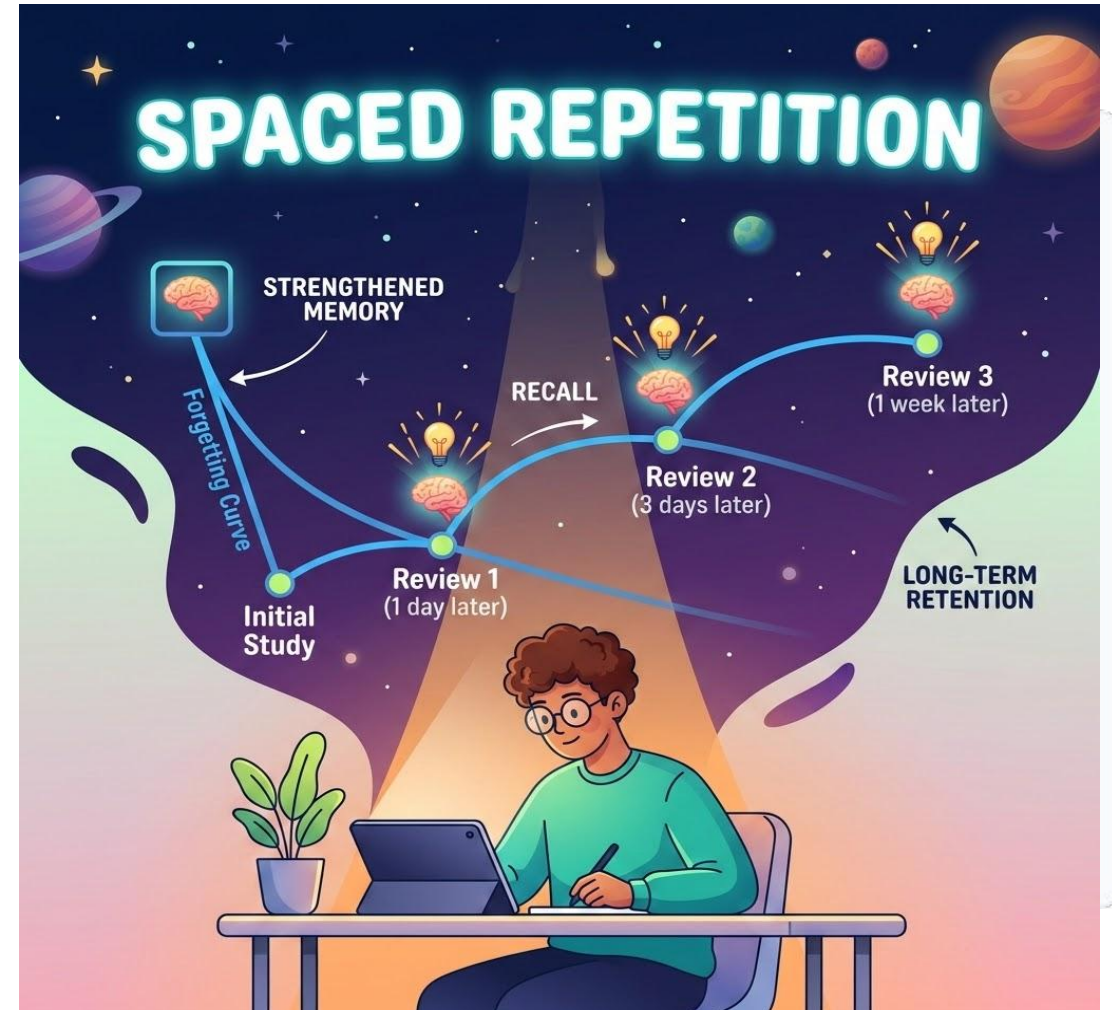
The Push for Mandatory Japanese

A formal **language requirement for PR** is being introduced via a mandatory **"Japanese Language and Lifestyle Learning Programme"**.

With **JLPT N2** as the discussed benchmark, the program will test skills and integration. Guidelines are expected by **FY2027**.

How Do We Learn?

- 🧠 **The Core Mechanic:** Psychologists have proven that reviewing information at dynamically expanding intervals halts the forgetting curve.
- 📅 **The Algorithm:** There's been a decade-long algorithmic competition to find the right interval: SM2 (1987), Free Spaced Repetition Scheduler (2024).
- ∞ **Permafrost Memory:** Some studies show recall spanning 50 years. Human memory forms deep, unshakeable roots under the right conditions, for some reason (?).





**Of course, we have the
most popular app and
influential program for this**







The beginning

JLPT N5 - N4 (2016-2018)

The Beginning: JLPT N5 & N4 (2016–2017)

JLPT N5: The Foundation

2016 • 4 Months Race

The easiest level is mostly focused on **symbol recognition**, including hiragana and katakana.

100 kanjis + grammar + listening

JLPT N4: Grammar

Jan 2017 – June 2017

Transitioning from isolated symbols to **contextual grammar understanding**.

Years of passive anime exposure transitioned from simple noise to a structured database of recognizable concepts.



JLPT N3: The Engineering problem

June 2017 – December 2017 The Context Deficit Challenge

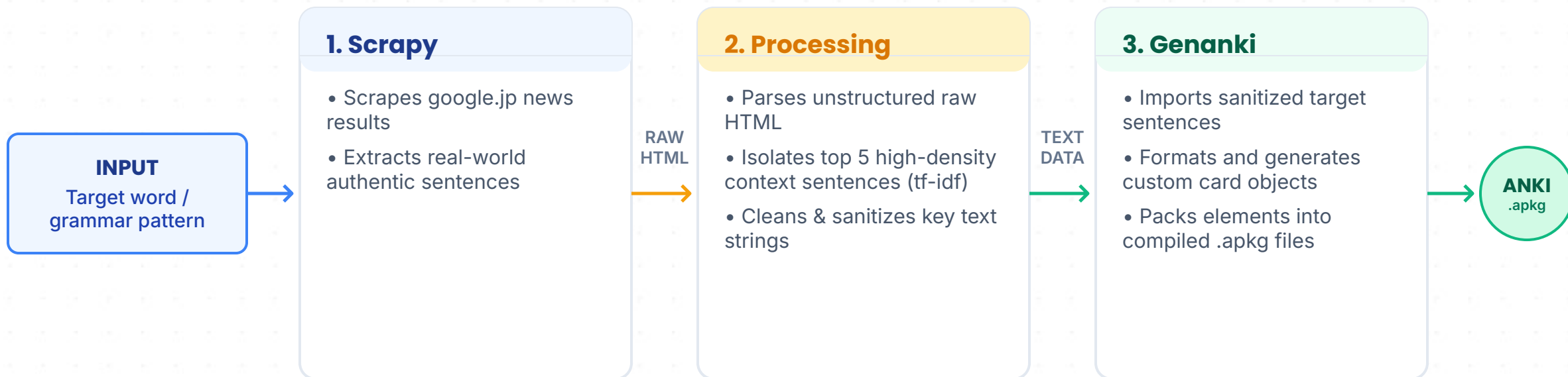
The leap in volume was immense—equivalent to **N5 and N4 combined** (500 characters & 1000 words).

Separated from Japan, I lacked the natural environment to apply this vocabulary. **How do you scale context when you have no one to speak to?**

Enter Python



JLPT N3: Adding context





And I stopped



2018 – 2019

2018 – The Gap Year & Pivot

I spent the year working and preparing a high-stakes MEXT scholarship application.

While the MEXT scholarship application ultimately failed, my previous tutor connected me directly with a Japanese CEO.

I secured an interview—this was the defining moment when **N3 fluency paid off, allowing me to speak and make myself understood.**



Japan



My first job (Sep 2019)

Not good enough

- N3 knowledge was good enough for communication, but not for a fully Japanese environment.
- I figured I had to get the N2 to get even more words to my brain.
- I lacked technical words (*yogo*) required for writing specifications and discussing with product managers.



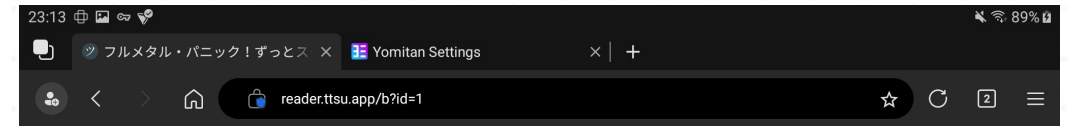


JLPT N2 (Jan 2020 – Dec 2020)

Immersion for N2 (Jan 2020–Dec 2020)

N2 brings **long lectures, specialized grammar**, 1500 words and 1000 Kanjis.

- My old pipeline was working fine for kanjis and grammar but still it was hard to maintain. so I tried the community supported add on **Yomichan**. It would generate Anki flashcards for any content I would see on the web.
- Being immersed tremendously helped especially on the reading and listening sections.
- In the professional aspect, I would learn a lot of **technical stuff..**



えいご 英語

CC100 526 VN Freq 3521 JPDF 4620.104458
Anime & J-drama 2662 BCCWJ-LUW 912
アクセント辞典 えいご [0] ●●●

デジタル大辞典 えいご【英語】

インド・ヨーロッパ語族のゲルマン語派に属する言語。現在、大ブリテンと北アイルランドのほか、アメリカ・カナダ・インド・オーストラリア・香港(ホンコン)などで使われ、国際的な傾向が強くなっている。語彙の面で、フランス語からの影響が大きく、ドイツ語・オランダ語などと親近関係にある。この語族の言語の中では、最も孤立語的な傾向を示す。

三省堂国語辞典 第八版 **えいご【英語】《名》

① イギリス・アメリカをはじめとした地域で話される、勢力の大きい言語。
② 「英語①」を学ぶ教科。英語科。

い いか かつ
め、ら ない。朝 全 して、ん いく
あ の 風 日 国 境 国 境 国 境
きる 若者の 人生を決めないで 欲しい! ...と 信二は
つくづく 思う。

特に源氏物語! あんな、一〇〇〇年前の恋愛至上主義の女が書いた妄想——ヤリチン男の腐りきった女遍歴なんぞが理解できるかどうかなかで、現代に生きる若者の人生を決めないで欲しい! ...と信二はつくづく思う。

文字が並
あったり
容は変わ
切れを眺

プロローグ



And I stopped

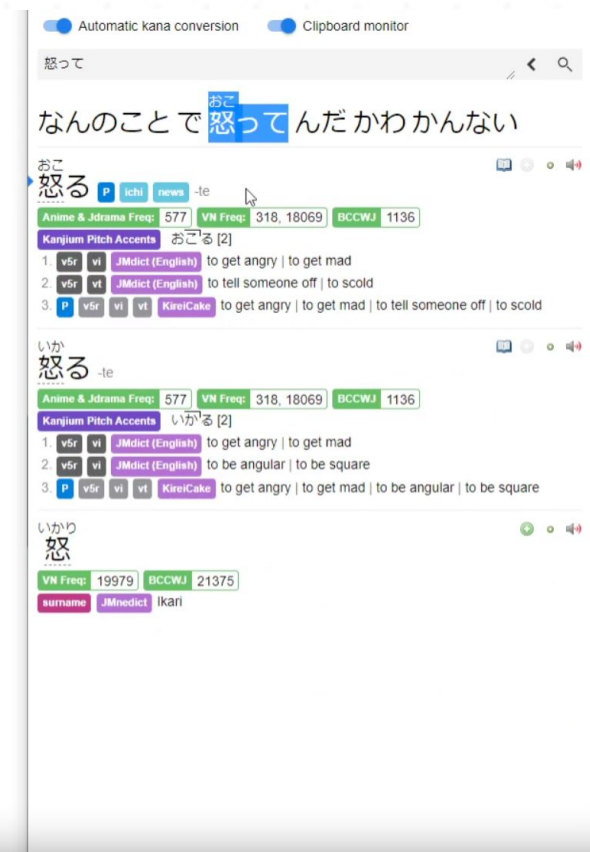


2021 – 2023 (The community moves forward)

Manga-OCR & Transformers

The Python community introduced [manga-ocr](#). Built on a Vision Encoder-Decoder transformer framework, it accurately parses vertical text, furigana, and stylized fonts from images, seamlessly linking native reading to flashcard creation.

The dataset was provided from a compilation of free mangas by the University of Tokyo (manga-109 dataset)



And chappy arrived



ChatGPT

Conquering JLPT N1 (Jan 2024 – July 2024)

Vocabulary & Kanji

Anki Was the Boss

Memorized around 1600 from the **2,136 Joyo Kanji**.

I had less time for vocabulary, but generated context-rich sentences with ChatGPT.

Grammar Synthesis

ChatGPT Integration

Python Selenium + ChatGPT.

I would run a script with genanki to generate the flashcards.

Comprehension

Organic Reading & Listening

Trusted the permafrost and mostly studied from books

December 2024 (The end)

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So it was over...

A close-up of a green cartoon owl's face. The owl has large, white, almond-shaped eyes with black pupils. Its beak is orange and slightly curved. The owl's face is a vibrant green color with a slight gradient. The background is dark green.

Not really

The illusion of mastery



The Recall Deficit

I had great recognition for kanji characters when reading, but froze when forced to produce them manually (writing unknown names, letters, etc.).

The Pronunciation Blindspot

Despite continuous speaking, I remained entirely oblivious to the fundamental importance of pitch accents.

My partner would constantly point out at my accent mistakes 😊



**Do you need a
Japanese waifu to
pronounce better?**

The Pronunciation gap



Not really

Onsei

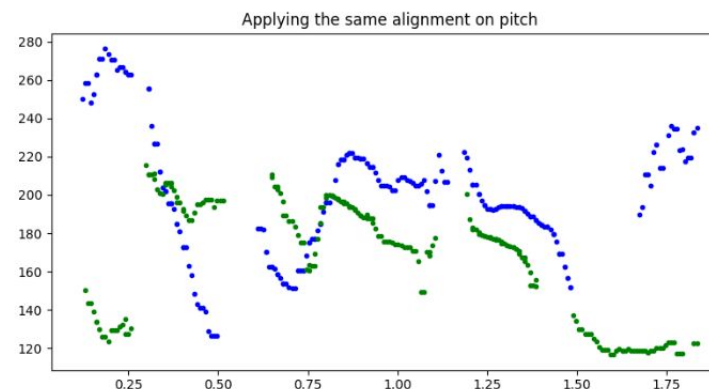
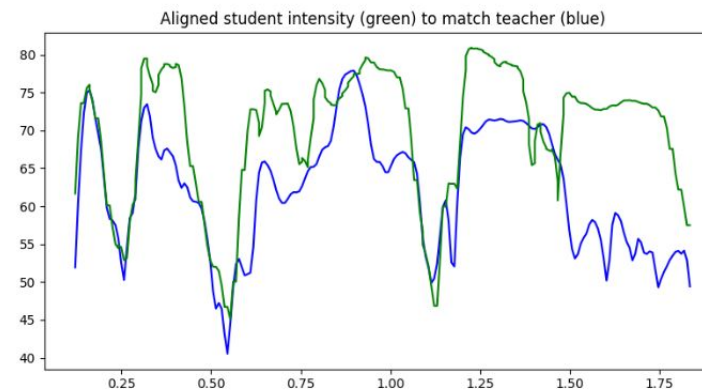
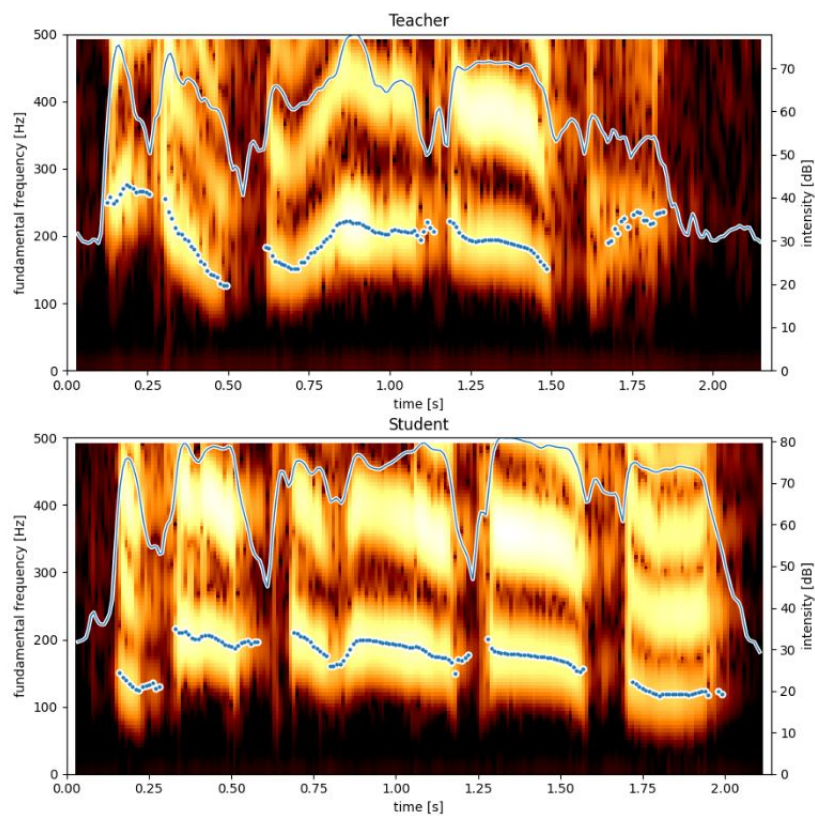
An ongoing project. It records your voice, analyzes the pitch contour, and grades your cadence against the standard Tokyo dialect.

NHK Pronunciation Dictionary

My gold standard. It provides professional announcer's recordings for every word.

Onsei

<https://github.com/itsupera/onsei>



The 2025/2026 Learning Stack



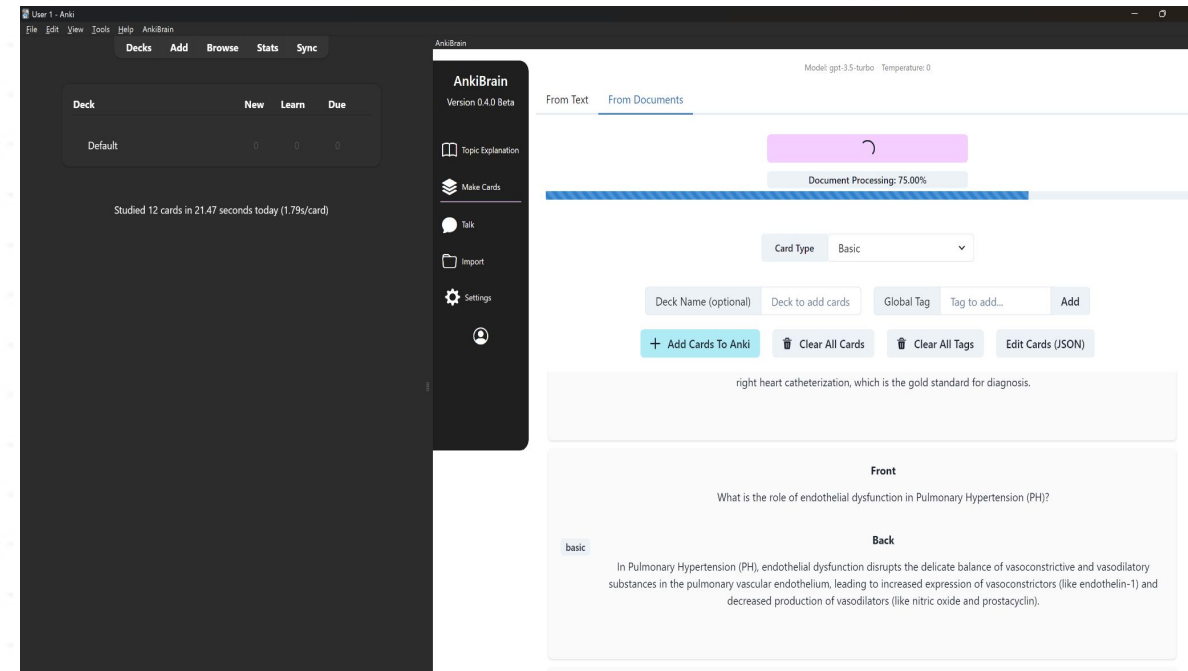
Claude Code & Genanki: Instead of simple scrapers, use agentic LLMs to parse specific datasets (like industry buzzwords) and automatically construct `.apkg` files.



Anki Brain: Advanced integrations now allow you to feed entire PDFs, technical ePubs, or corporate documents into an AI, extracting the vocabulary instantly.



Organic Returns: Eventually, the goal is to step away from the flashcards and let the language become second nature through pure consumption.



A never-ending journey

- ✓ **Immerse yourself:** Aim for organic learning. It's more fun that way.
- 👂 **Recognition vs Recall. :** The tests doesn't teach you pronunciation and writing.
- 📈 **The Professional Multiplier:** Having reliable, business-level Japanese can boost your career: broad interview access, negotiation edge, and more friends :)



**Will this matter in
50 years?**

I believe so. Language is your ultimate insurance.

Questions?

Thank you for listening!

